Team Decode

Neuro Vision

AI for a World Beyond Sight

AI-powered assistance for navigating, reading and understanding the world

The Problem

The world communicates through sight

The Visual Gap

Everyday information obstacles, signs, labels, directions and surroundings is primarily communicated visually

The Cost of Dependence

Simple tasks can require assistance: reading a sign, identifying an object, navigating a new place or knowing what lies ahead

The Reality

They don't need someone to see for them. They need a way to understand the world independently

The Solution

What it is

An AI-powered personal assistant built for visually impaired individuals

Neuro Vision combines AI vision, voice interaction, navigation and emergency assistance into one smartphone experience

How it works

See → Speak → Navigate → Assist

- AI Vision understands surroundings and reads text.
- Voice Assistant lets the user interact hands-free.
- Navigation guides the user to a destination.
- SOS connects the user to a saved emergency contact when they need help.

The Result

A single device that helps users understand what is around them, read information, find their way, and reach someone they trust when they are stuck

How It Works (Under the Hood)

Architecture Designed for Action.

Perception

CameraX captures the surroundings while YOLO + ONNX detects objects and ML Kit reads visible text. GPS provide additional context

Intelligence Engine

The app processes detections, text, voice input and location to understand what matters and prioritise useful information for the user

Guidance & Assistance

Text-to-Speech converts insights into voice guidance, while navigation provides directions and SOS connects the user to a saved emergency contact

The Accessibility Advantage

Why NeuroVision goes beyond individual accessibility tools

Integrated Accessibility

Vision, OCR, navigation, voice and SOS in one experience

Real-Time Assistance

On-device YOLO + ONNX processes the camera feed to detect surrounding objects and provide immediate voice guidance.

Designed for Independence

Helping users read, navigate, understand, and reach out for help without constantly relying on another person

Future Scope

From an AI app to a personalized accessibility ecosystem

Adaptive Intelligence

Train models for new objects, hazards, environments, and accessibility use cases

Wearable & Spatial Intelligence

Integrate smart glasses and build personalized maps of homes and frequently travelled paths

Connected Accessibility

Connect smart canes, NFC-enabled objects, and other assistive devices into one ecosystem

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Thank You